Physical Chemistry Cume December 4th 2015

This cume refers to the following announced publication:

Water desalination with a single-layer MoS_2 nanopore. Mohammad Heiranian, Amir Barati Farimani & Narayana R. Aluru. Nature Communications, vol 6, article 8616 (6 pages + Supplementary info).

1) What is the rationale for the work? Describe the current state-of-the-art of water purification and the hurdles that the technology described in the paper hopes to overcome. (30 points)

2) The key results from the paper are shown in figure 2 (below, annotated to allow b/w printing)

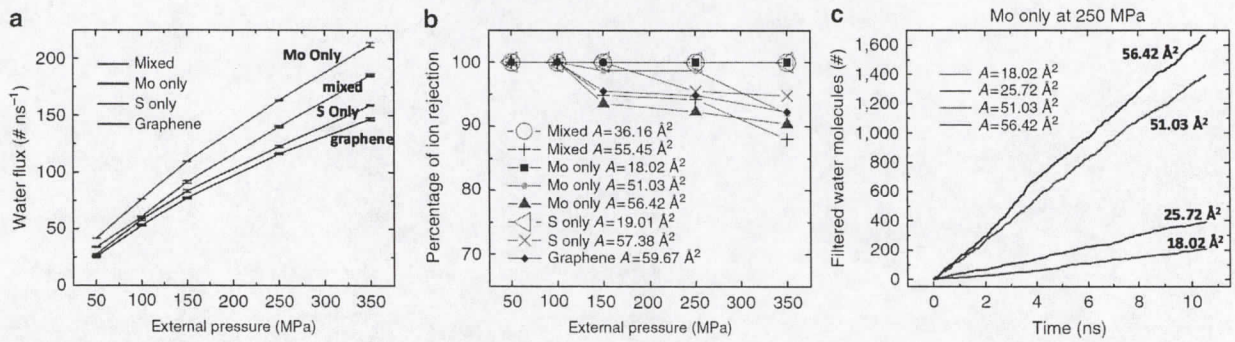

Figure 2 | Water permeation and salt rejection. (**a**) Water flux as a function of the applied pressure for mixed, Mo only, S only and graphene nanopores with similar pore areas. (**b**) Percentage of ion rejection by various pores as a function of the applied pressure. Pores with different edge chemistries as well as various pore areas (denoted by *A*) are considered. (**c**) Number of water molecules (#) filtered through Mo only pores as a function of simulation time for different pore areas at a fixed pressure of 250 MPa.

Describe the physical chemistry and geometry considerations underlying the following observations: (10 points each)

a) At a given external pressure, the Mo only nanopores show a higher water flux than the mixed or S only nanopores.
b) MoS_2 shows a higher water flux than graphene, independent of whether the nanopore is Mo only, S only or mixed.
c) The ion rejection percentage does not depend on pressure for small pores, but does for larger pores.
d) For the pores with equivalent areas, ion rejection is quite similar regardless of the type of pore.
e) The rate of filtered molecules at a given pressure increases with nanopore area.

3) Using the data in figure 2, how many Mo only nanopores would be needed in a single membrane to filter 100 ml of water in 10 s at 250 MPa pressure if the pore size is a) 56.42 Å^2 or b) 18.02 Å^2. What would be the final NaCl concentration in each case if you started with 1M NaCl. Take the density of water to be 1.00 g/ml. (20 points)